UNIVERSITÀ DI TRENTO

Exploring data from turning experiments

Report

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In this analysis of CNC turning experiments, key parameters were derived for two machining configurations: one at constant cutting speed (V_c) and the other at constant spindle speed (N). Based on the measured spindle power, the cutting power (P_c), cutting force (F_c), and specific cutting force (K_c) were calculated for each working diameter. To achieve this, the data were filtered to isolate only the segments corresponding to the effective cutting phases.

The machining parameters were:

- Work material: AISI 4135
- Diameter reduction (X-axis): from 48 mm to 24 mm (measurements from 46 mm onward)
- Machining length (Z-axis): 40 mm
- Depth of cut: $a_p = 1$ mm
- Feed rate: $f = 0.2$ mm/rev
- Constant cutting speed: $V_c = 150$ m/min
- Constant spindle speed: $N = 1200$ rpm

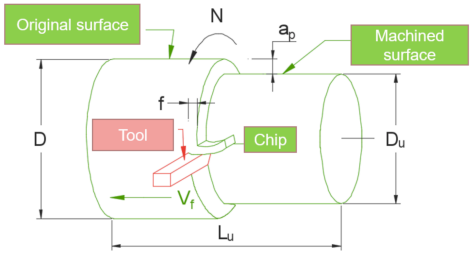


Figure 1.1. Turning setup used in the experiments.